

Statistical Shape Analysis of Brain Arterial Networks: Generative Extension

MVA Project - Geometric Data Analysis

Enzo Azzoug
MVA - ENS Paris-Saclay
Paris, France
enzo.azzoug@eleves.enpc.fr

Tanguy Lourme
MVA - ENS Paris-Saclay
Paris, France
tanguy.lourme@eleves.enpc.fr

Abstract

This report provides a critical analysis and rigorous extension of the framework proposed by Guo et al. [2] for the statistical analysis of Brain Arterial Networks (BANs). We first detail the theoretical foundations of the method, justifying the necessity of a Riemannian quotient space approach over traditional Euclidean statistics. We deeply analyze the mathematical construction of elastic graphs, the role of group actions in handling nuisance variables, and the properties of the metric. Beyond the descriptive analysis provided in the original paper, we introduce one major contributions: a probabilistic generative model based on a multivariate Gaussian distribution defined on the tangent space of the shape manifold (code). Finally, we discuss the numerical and topological limitations of the current graph matching approach.

Keywords

Elastic Graphs, Riemannian Geometry, Quotient Space, Generative AI, Tangent PCA

1 Context and Motivation

1.1 Why Standard Statistics Fail

Brain Arterial Networks (BANs) are critical anatomical structures for understanding neurovascular pathologies (aneurysms, strokes) and aging. However, they present a fundamental challenge for statistical analysis: they are non-Euclidean objects with varying topology as we can illustrate in 1. Standard statistical methods (PCA, Euclidean Mean) rely on the assumption that observations belong to a vector space $\mathbb{R}^d$ of fixed dimension d , where components are ordered and comparable. BANs violate these assumptions on three levels:

- (1) Variable Dimensions: Different subjects have different numbers of arteries and bifurcations. A vector representation would require an arbitrary cutoff or zero-padding that destroys geometric consistency.
- (2) Unordered Structure: Graphs are invariant to node permutation. If we represent a graph by an adjacency matrix, swapping rows/columns represents the *same* object, but the Euclidean distance between these matrices would be non-zero.
- (3) Non-Linear Geometry: The shape of an artery is a curve in 3D space. Linear averaging of 3D coordinates destroys the curvature constraints (e.g., the average of two C-shapes might be a straight line, which is anatomically physically impossible).

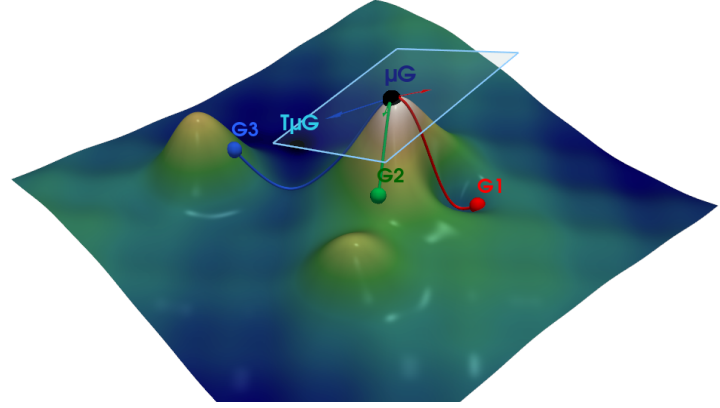


Figure 1: Visualization on a Riemannian manifold $\mathcal{G}$. We observe the tangent space $T_{\mu_G}\mathcal{G}$ at μ_G . The vectors in the tangent space define initial speed vectors for geodesic shooting from μ_G to G_1, G_2, G_3 .

1.2 Target Applications

The primary goal of this framework is to develop a "Shape Space" for BANs to enable:

- Comparing patients: Quantifying how far a specific patient is from another one. This can be used to build clustering algorithms or anomaly detection.
- Atlas Construction: Computing a statistically robust "average brain network" (μ) for a population.
- Variance Analysis: Understanding how factors like Age or Gender deform the geometry of the network (Geodesic Regression).

1.3 Alternative Methods and Limitations

Prior to Guo et al. [2], two main approaches existed:

- Feature-based Analysis: Extracting scalar features (total length, number of loops, average tortuosity) and performing stats on these vectors. *Limitation:* This is a lossy projection. It is impossible to reconstruct the organ from these scalars, and local geometric information is lost since we project on a space of lower dimension.

- Topological Data Analysis (TDA): Using persistent homology to compare topological features (loops, holes). *Limitation*: TDA is excellent for classification but poor for registration. It does not provide a point-to-point correspondence between arteries, which is required for surgical planning or local deformation analysis.

2 Mathematical Formalism

The paper proposes a rigorous Riemannian framework to solve the issues mentioned above. We analyze each component of this framework below.

2.1 The Edge Space: SRVF Representation

Each artery is modeled as a parameterized curve $\beta : [0, 1] \rightarrow \mathbb{R}^3$. To compare shapes properly, the metric must be invariant to re-parameterization (the speed at which we traverse the curve shouldn't change its shape). The authors use the Square Root Velocity Function (SRVF) transform [3]:

$$q(t) = \frac{\dot{\beta}(t)}{\sqrt{|\dot{\beta}(t)|}}. \quad (1)$$

Why this transform? The standard L^2 distance on coordinates β is not re-parameterization invariant. The L^2 distance on q functions, however, corresponds to a specific elastic metric on β that allows stretching and bending. This transforms the complex non-linear shape space into a subspace in a Hilbert space L^2 , simplifying computations significantly. Moreover, working on $\dot{\beta}(t)$ instead of β simplifies the registration problem since this transformation is translation-invariant.

Then they introduce the action of a diffeomorphic group. Let $\gamma : [0, 1] \rightarrow [0, 1]$ a diffeomorphism then $q * \gamma = (q \circ \gamma) \sqrt{\dot{\gamma}}$ that is actually the SRVF of $\beta \circ \gamma$. Then we can define classes $[q] = \{q * \gamma | \gamma \in \Gamma\}$. With $\mathcal{S}$ the set of classes we can define a metric to compare two curves: $d_S([q_1], [q_2]) = \inf_{\gamma} \|q_1 - (q_2 * \gamma)\|$. This metric allows us to compare two curves without caring about their parametrizations. If one is the same as the other after re-parametrization then the distance between both will be equal to zero. After the matching step it allows to work with one representant per class.

2.2 The Elastic Graph Representation

A BAN is represented as an attributed graph $G = (A, u)$.

- $A \in \mathcal{S}^{n \times n}$: A symmetric adjacency matrix where each non-zero entry $A_{ij} = [q_{ij}]$ is the class of the SRVF of the artery connecting node i and j .
- $u \in \mathbb{R}^n$: A vector of node attributes (e.g., degree).

The space of such graphs is denoted $\mathcal{B}$. Then, naturally to compare two graph we need to compare both matrices and both vectors. For the matrices they inspire from the Frobenius norm plugging in the distance defined below $d_a(A_1, A_2) = \sqrt{\sum_{i,j} d_S^2(a_{i,j}^1, a_{i,j}^2)}$. And for the edges they introduce $D = |u_1 \mathbf{1}_n^T - \mathbf{1}_n u_2^T|$. Finally the distance on $\mathcal{B}$ is the weighted sum $d_b = d_a + \lambda \text{Tr} D$.

2.3 The Quotient Space Construction

The raw space $\mathcal{B}$ is not the true shape space because of equivalent transformations. Two graphs should be considered identical if they differ only by labeling or rigid motion. The authors define the shape space $\mathcal{G}$ as a quotient space:

$$\mathcal{G} = \mathcal{B} / (\mathcal{P} \times SO(3)). \quad (2)$$

2.3.1 Group Action 1: Permutation $\mathcal{P}$. The group of permutation matrices acts on the adjacency matrix by $P \cdot A = PAP^T$ and on nodes by $P \cdot u = Pu$. To delete the differences caused by an arbitrary labeling of the indexes of the nodes we really need to use this group action to be able to compare graphs in a proper way. This introduces the *Graph Matching* problem [4] into the distance calculation.

2.3.2 Group Action 2: Rotation $SO(3)$. The rotation group acts on the curves (edges) and apply the rotation $O \in SO(3)$ to all the SRVF classes. This allows not to be sensitive to the rotations when comparing two curves. In a medical context it is very useful since two patients never have the same exact position when performing an MRI for example. If we add this to the translation invariance mentioned before, we reach rigid motion invariance (translation + rotation).

2.4 The Metric and Hyperparameters

The geodesic distance between two graph shapes $[G_1]$ and $[G_2]$ in $\mathcal{G}$ is defined by minimizing the distance in the total space over the group actions:

$$\begin{aligned} d_{\mathcal{G}}([G_1], [G_2]) &= \inf_{\substack{P \in \mathcal{P} \\ O \in SO(3)}} d_b\left((A_1, u_1), (O \cdot PA_2P^T, Pu_2)\right) \\ &= \inf_{\substack{P \in \mathcal{P} \\ O \in SO(3)}} \underbrace{\left(\sum_{i,j} d_S^2(q_{ij}^1, O \cdot q_{P(i)P(j)}^2) \right)^{1/2}}_{\text{Edge Term}} \\ &\quad + \underbrace{\lambda \sum_i |u_{1,i} - u_{2,P(i)}|}_{\text{Node Term}}. \end{aligned} \quad (3)$$

Analysis of λ : The hyperparameter λ balances the importance of geometric structure (edges) vs. graph structure (nodes).

- **Limit $\lambda \rightarrow 0$:** The matching is driven solely by edge shapes. A central artery connected to many others could be matched with an artery with the same shape but far from the center and with less connectivity.
- **Limit $\lambda \rightarrow \infty$:** The problem reduces to a simple graph registration. The curves are ignored so a short and straight artery could be matched with a long and looping artery if their respective nodes share the same degrees.

2.5 Statistics on the graph space

Now that they provided a distance and geodesics in that space we are able to compute the Fréchet mean : an element (not always unique) in the space that minimizes the sum of the geodesic path lengths to the others. And once the Fréchet mean is computed, it is possible to do a Tangent PCA. First, all graphs should be registered.

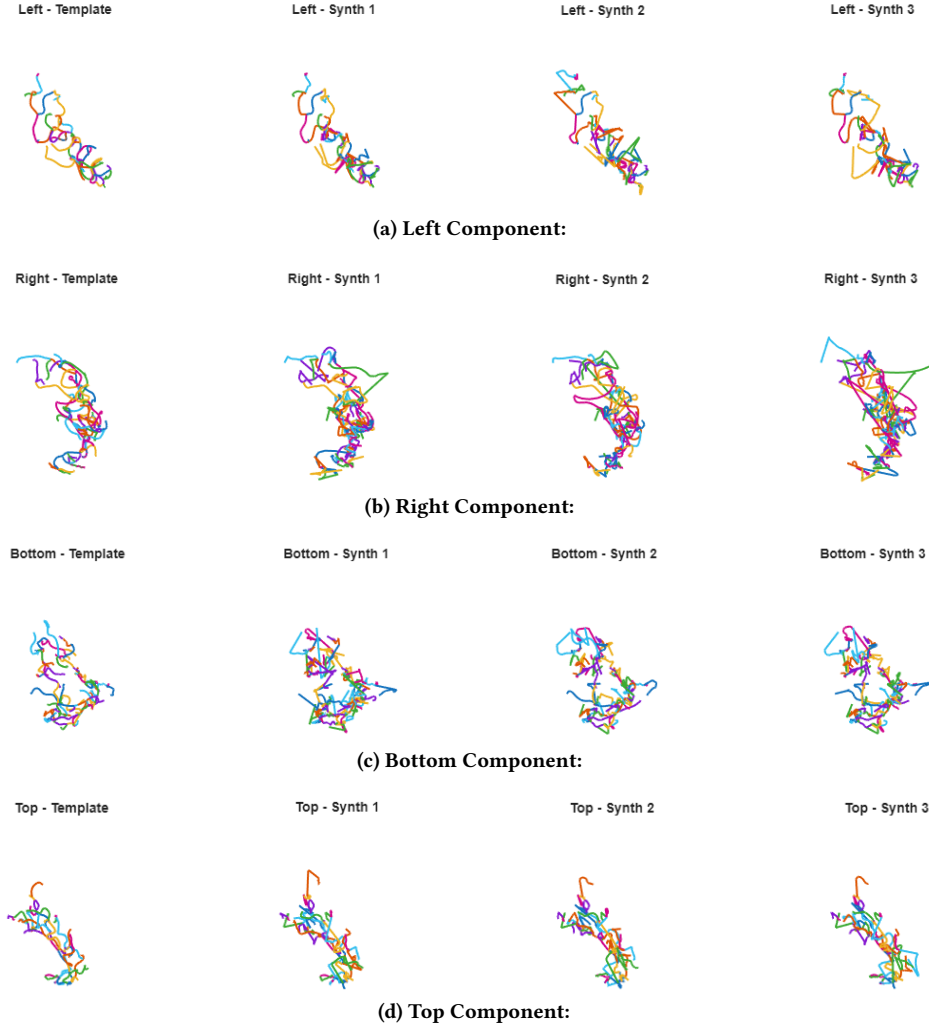


Figure 2: Generative Results across all anatomical components. Each row compares a Real Template (Col 1) with three Synthetic Samples (Cols 2-4). The model successfully captures component-specific geometric features (curvature, scale, density) while maintaining the topological constraints of the template.

Then, we compute the difference (Log) with the mean as we do for standard PCA. Locally, the space is euclidean and we have shooting vectors that we can remap to the original registered graphs (Exp). And on this euclidean space we can perform statistics and PCA as we are used to.

3 Novel Contributions and Methods

While the original paper focuses on *descriptive* statistics (Fréchet mean, Tangent PCA). Our approach now is to use these statistical tools to apply what we can do on euclidean spaces in this particular case.

3.1 Data augmentation via Tangent Space

3.1.1 Objective. In general in medical domain it is difficult to have a high quantity of data for many reasons : rare diseases, privacy...

So it is impossible to use machine learning or deep learning models that learn thousands or millions of parameters with only hundred examples. And using data augmentation can solve this problem while providing synthetic examples that we can generate as much as we want. The main issue is to generate good and plausible examples. We need synthetic data that has the same statistical properties as the original dataset : shapes, graph topology. Thanks to their contribution we are now able to project on the tangent space (registration + difference with the mean) to do statistics on an euclidean space and then come back to the original graph space by an exponential shooting.

3.1.2 Mathematical Formulation. To generate plausible synthetic BANs, we cannot simply average node coordinates due to the non-linear nature of the shape space. Instead, we exploit the local linear

structure of the tangent space at the Fréchet mean μ .

Let $\mu = (A_\mu, u_\mu) \in \mathcal{G}$ be the mean graph (in the sense of Fréchet mean). We use the logarithmic map $\text{Log}_\mu : \mathcal{G} \rightarrow T_\mu \mathcal{G}$ which maps a graph G_i to a tangent vector $v_i \in T_\mu \mathcal{G}$. Since the graphs are aligned via P_i and O_i to get $G_i^* = (A_i^*, u_i^*)$ during mean computation, this simplifies to the difference in the SRVF space. So Log_μ first registers G_i with μ and then computes the difference in the Hilbert space:

$$v_i = A_i^* - A_\mu \in L^2(\mathbb{R}^3)^{n^2}. \quad (4)$$

Where n is the number of nodes. We assume that the population of tangent vectors $\{v_i\}_{i=1}^N$ follows a multivariate Gaussian distribution $\mathcal{N}(0, \Sigma)$ on the tangent bundle. We estimate Σ via PCA:

$$\Sigma \approx \frac{1}{N-1} V V^T, \quad \text{where } V = [v_1, \dots, v_N]. \quad (5)$$

To generate a new sample, we draw $z \sim \mathcal{N}(0, I)$, compute $v_{\text{new}} = U \Lambda^{1/2} z$ (where U, Λ are eigenpairs of Σ obtained through SVD), and map it back to the manifold using the exponential map:

$$A_{\text{new}} = \text{Exp}_{A_\mu}(v_{\text{new}}) = A_\mu + v_{\text{new}}. \quad (6)$$

However, this formulation is incomplete as it strictly focuses on the edge shapes A while omitting the node attributes u_i^* because in their model they always pad with zeros so it is equivalent to register all the nodes to the largest graph (the one with the larger amount of nodes). To address this problem we need to select a graph as a template to receive the curves that are generated through the gaussian. As they do, we represent a curve with a finite dimension vector using a grid so we can consider $v \in \mathbb{R}^{3 \times n^2 \times p}$ (3 dimensions for x, y, z , n^2 nodes and p values for the grid). Let's note X_G the uniform random variable on the (discrete) space of graph examples and Y_G the random variable associated to the sampled plausible graphs. To sample Y_G we need to condition with X_G :

$$\mathbb{E}[Y_G | X_G] = f(X_G, A_\mu, v_{\text{new}}). \quad (7)$$

Where $f : \mathcal{G} \times \mathcal{G} \times \mathbb{R}^{3 \times n^2 \times p} \rightarrow \mathcal{G}$ selects the nodes and edges using the template X_G and does the integration and applies a rigid transform to β for each edge to match with the starting and ending node. If we don't apply such a transformation it will lead to a mess of curves starting at $(0, 0, 0)$ and going in every direction which would not be plausible. So thanks to their work we are able to sample a multivariate gaussian and a template which corresponds to a discrete sampling and propagate this through a deterministic function f to obtain plausible synthetic data. We could also add more "diversity" on a visual aspect by composing with permutation and rotation matrices. But actually this would not be pertinent since it would represent the same element in $\mathcal{G}$ and would need another matching step to compare with others.

3.2 Experimental Results

We validated our generative framework by training the model on the four main components of the Brain Arterial Networks : Left, Right,

Top, Bottom (one model for each component). For each component, we computed the Fréchet mean and the covariance matrix in the tangent space. Due to computational costs we only did it for 13 samples for each component : 1 template + 4×3 because we could use 4 workers in parallel. To generate synthetic samples, we sampled tangent vectors from the learned multivariate Gaussian distribution and mapped them back to the manifold using the exponential map, applied to a randomly selected template topology from the dataset.

As illustrated in Figure 2 the synthetic BANs exhibit realistic geometric properties.

- **Geometric Plausibility:** The generated arteries display smooth curves with tortuosity levels consistent with real anatomical data. The model does not produce "broken" or degenerate lines, validating the use of the SRVF framework which ensures smooth geodesics.
- **Structural Coherence:** By projecting the Gaussian variations onto a fixed template topology (the "Real Template" shown in the first column of each figure), we ensure that the connectivity of the network is preserved. For instance, in all of the components, we can clearly track how specific vessels deform (widen, contract, or bend) across the three synthetic examples while remaining attached to the correct nodes.
- **Diversity:** We managed to keep the global structure and shape for every component but each sample is different. Indeed, we observe arteries that vary significantly in length and curvature compared to the template. It allows to model the existing diversity across patients. Of course we were constrained by computational costs and by training with the full dataset we would observe even more plausible outputs. As an example we could observe the specificity of a patient A transported on the arterial structure of a patient B.

At a first sight it looks very similar with the original dataset. We tried to analyze it a bit more precisely not just saying "it looks like the real data". So we computed both length and curvatures for the curves. For each real or synthetic patient, we computed the mean and we compared the distribution that corresponds to the 92 real patients with the distribution obtained with 92 "synthetic" patient but still with 13 training examples. In Figure 3 we can see that for the mean curvature the distribution of the synthetic dataset fits well with the original distribution. One other interesting point is that even if the training dataset is not representative of this (not the same mean) the model fits well for the curvature. We can conclude that working on the tangent space allows to model local shapes. Then, if we look at the average length we don't fit with the original distribution. Both distributions look like gaussian distribution but then mean value is not the same and the variance seems also different. A reason behind this can be the last step in our model when we use the exponential map. As a reminder once we get the sample from the multivariate gaussian we obtain the different values for the selected template's adjacency matrix and we need to integrate the SRVF to get the curves. But as we mentioned earlier we force the starting and ending point to be the good nodes to keep a plausible structure and this can stretch the curves modifying their length but not their curvatures.

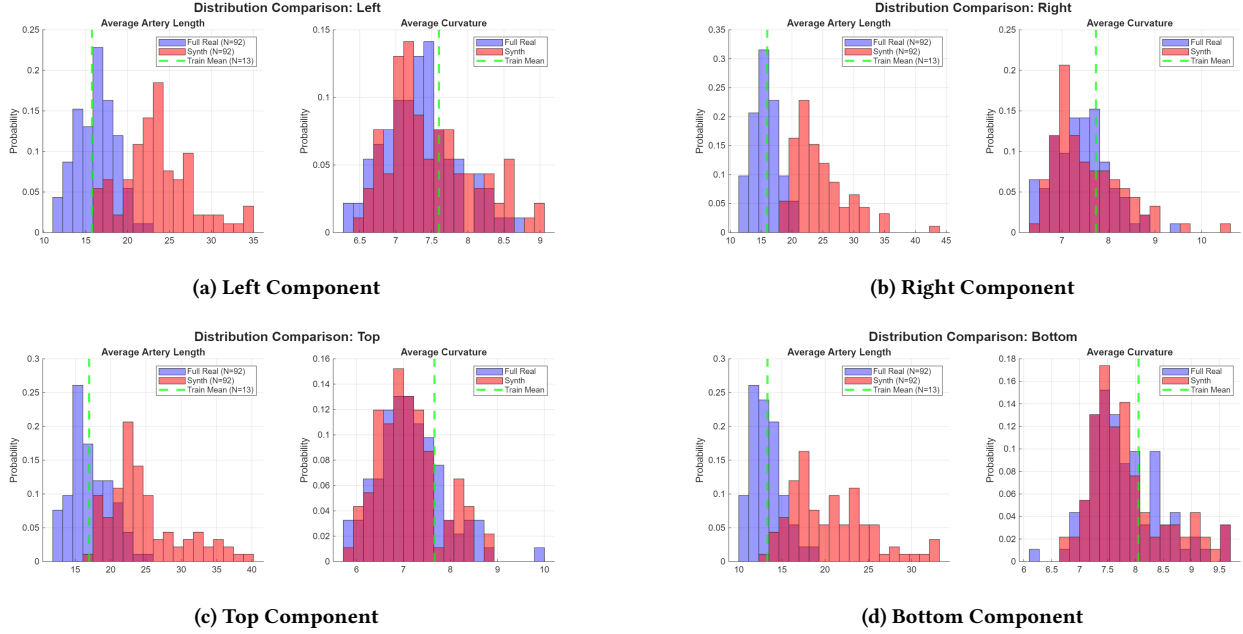


Figure 3: Distribution of length and curvatures for all of the 4 components for real data (Blue, $N = 92$) and synthetic data (Red, $N = 92$). The green vertical line corresponds to the value of the mean on the training dataset ($N = 13$).

An idea for future work would be to introduce in this work the use of optimal transport in order to fit with such distributions that describe the best the networks. It could also be a way to generate smoother curves because sometimes curves angles are a bit abrupt due to representation in the discret domain.

4 Critical Analysis and Limitations

4.1 The Topological Alignment Problem

The fundamental limitation of this framework is the assumption that all BANs can be mapped to a "super-graph" via null-node padding.

- **Biological Reality:** Arteries are not just "present or absent" (null edges); bifurcations can vary topologically. The paper handles this by forcing a matching, sometimes leading to topologically distinct graphs being compared solely by geometry.
- **Ghost Features:** The use of null edges means the mean shape μ depends heavily on the largest graph in the dataset. If one outlier has a unique, rare vessel, every other graph in the dataset must "hallucinate" this vessel as a null edge to perform statistics.

4.2 Computational Scalability

The graph matching step (FGM) is computationally expensive. The authors resort to approximating the mean by registering everything to the "largest graph" instead of performing pairwise matching for the true Fréchet mean. This breaks the theoretical guarantee of the Fréchet mean (it becomes a "Frechet mean of a subset"). For large-scale datasets this is a bottleneck.

4.3 Tangent Space Linearization

Our generative extension and the paper's PCA rely on the tangent space being a good approximation of the manifold. However, if the spread of BAN shapes is large (high curvature of $\mathcal{G}$), the exponential map Exp_μ may distort distances. A "Principal Geodesic Analysis" (PGA) [1] would be theoretically superior to Tangent PCA, though harder to implement. Indeed, with PGA we would need to compute much more geodesic distances and for that we must match the corresponding graphs which is extremely costly in this context.

4.4 Intrinsic Geodesic Regression

While Guo et al. analyze the effect of aging by performing linear regression on the Principal Component scores, this approach implicitly assumes that the manifold is flat enough to treat the tangent space at the mean as a global proxy. We propose a more rigorous approach: *Geodesic Regression*.

Instead of minimizing a Euclidean error on PCA scores, we model the evolution of the BAN shape $G(t)$ as a function of age t via a geodesic curve γ on the manifold $\mathcal{G}$. The model is:

$$G(t) = \text{Exp}_\mu(t \cdot V). \quad (8)$$

where μ is the geometric baseline (e.g., BAN at mean age), and $V \in T_\mu \mathcal{G}$ is a "velocity vector" representing the structural aging process.

To estimate (μ, V) , we minimize the sum of squared Riemannian distances:

$$E(\mu, V) = \sum_{i=1}^N d_{\mathcal{G}}^2(\text{Exp}_\mu((t_i - \bar{t}) \cdot V), G_i). \quad (9)$$

where G_i is the observation for patient i and t_i their age. Unlike the paper’s method which separates shape directions (PCA) from aging correlation, this method directly finds the "Aging Direction" in the shape space. If implemented, we expect V to capture the increase in tortuosity and variance mentioned in the paper’s conclusion.

5 Conclusion

In this report, we have analyzed and extended the Riemannian framework proposed by Guo et al. for the statistical shape analysis of Brain Arterial Networks. By modeling BANs as elastic graphs within the quotient space $\mathcal{G} = \mathcal{B}/(\mathcal{P} \times SO(3))$, this approach provides a rigorous mathematical foundation to perform statistics on objects that exhibit both geometric (tortuosity) and topological (branching) variability.

Our primary contribution was to exploit the local linearity of the tangent space at the Fréchet mean to formulate a geometric generative model. By sampling from a multivariate Gaussian distribution on the tangent bundle, we are able to synthesize plausible continuous deformations of the arterial shapes, effectively filling the voids in the shape manifold between observed subjects.

However, a strict limitation remains: our current model relies on a fixed graph template, meaning it generates geometric variations but cannot currently simulate topological changes (e.g., missing arteries or distinct bifurcation patterns). The matching process effectively locks the topology to that of the mean shape.

Consequently, the most promising direction for future work lies in a hybrid data augmentation strategy. By combining our tangent-space geometric sampling with a discrete probabilistic model for topological perturbations (such as graph edit operations consistent with anatomical priors), one could generate fully diverse synthetic datasets. Such a dual approach would constitute a complete solution to the "Small Data" problem, finally unlocking the potential of data-hungry Deep Learning models like GNNs for complex medical structures.

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